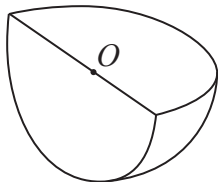


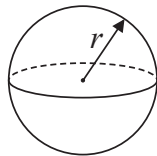
19 Shape **S** is one quarter of a solid sphere, centre O .



Shape **S**

Volume of sphere = $\frac{4}{3}\pi r^3$

Surface area of sphere = $4\pi r^2$



The volume of **S** is $576\pi \text{ cm}^3$

Find the surface area of **S**.

Give your answer correct to 3 significant figures.

You must show your working.

..... cm^2

(Total for Question 19 is 5 marks)